

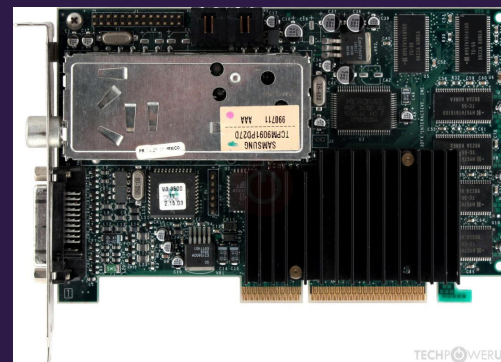


Thermal Behaviors in Liquid Immersion Cooling under Various Workloads: a Case Study

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Costs of Ineffective Cooling

- Energy expenditure → heat → performance reduction
 - Lower clock rate
 - System taken offline
 - Hardware damage
- Cooling systems maintain performance
 - Sustain activity
 - Recover faster
- Dense compute == higher performance AND thermal burden
 - GPUs and other specialized chips
 - Towards kW-scale TDP



Voodoo3 3500 TV AGP (1999, 15W TDP)



NVIDIA H100 (2024, 350W TDP)

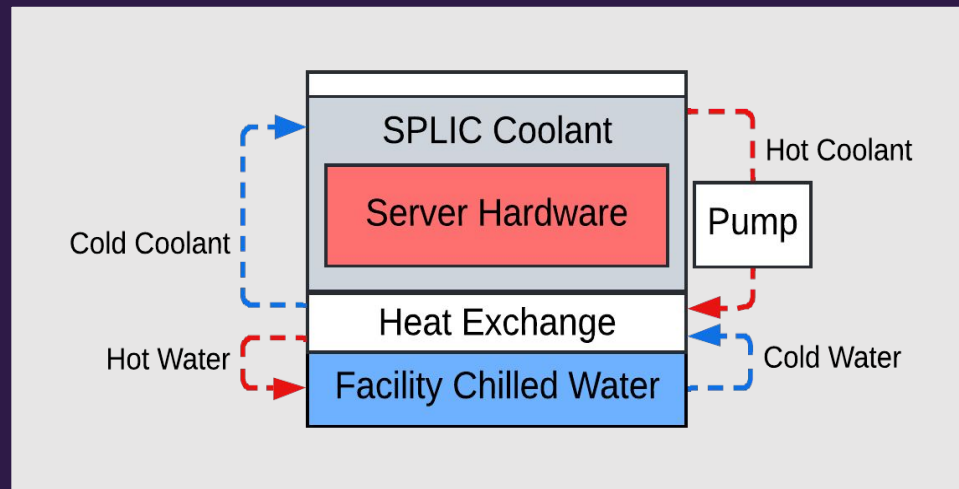
Available Techniques for Effective Cooling

- Heat dissipation via liquid > gas
 - Distilled water possible, but MUST remain pure!
 - Many dielectric fluids similar to water in viscosity and thermal conductivity
- Liquid cooling and evaporators well-established technologies, developing best integration with servers
 - Key future developments
 - Novel fluids
 - Liquid circulation practices
 - Specialized hardware for liquid immersion



Single Phase Liquid Immersion Cooling (SPLIC)

- Previous studies established
 - Foundational effectiveness of liquids over gases
 - Designs and properties for dielectric fluids
 - Placement and number of pumps in tank
 - Impacts on hardware warranties
- Some effects not studied
 - Separate sensors drive cooling systems
 - Delay in system's reaction
 - Varying activity and physical arrangements



≡ Case Study Contributions

- Understand thermal burdens by component
 - Commodity hardware is compatible, but not maximally effective
 - Potential to share heat between components
 - System cooled equally rather than per-component
- Observe heat dissipation capabilities under load
 - Determine appropriate system responsiveness
 - Other liquid cooling approaches can improve efficiency at high temperatures
- Differentiate heat accumulation based on real workloads
 - Hardware utilization
 - Periodicity
 - Demand



☰ Hardware Listing

- Cooling Hardware: Submer SmartPod v3
 - White mineral oil dielectric coolant
 - 11C datacenter facility water
- Server Hardware:
 - 3 distinct nodes with two architectures
 - (2 CPU nodes) Intel Xeon E5-2670 v3 2.3 GHz 64 cores
 - (1 GPU node) AMD EPYC 7551P 2.0 GHz 32 cores;
two NVIDIA Titan V 12 GB HBM2 GPUs
- Preparation for Immersion:
 - Thermal paste → indium foil
 - Remove fans / blades from power supplies, GPUs, CPUs



SC '24

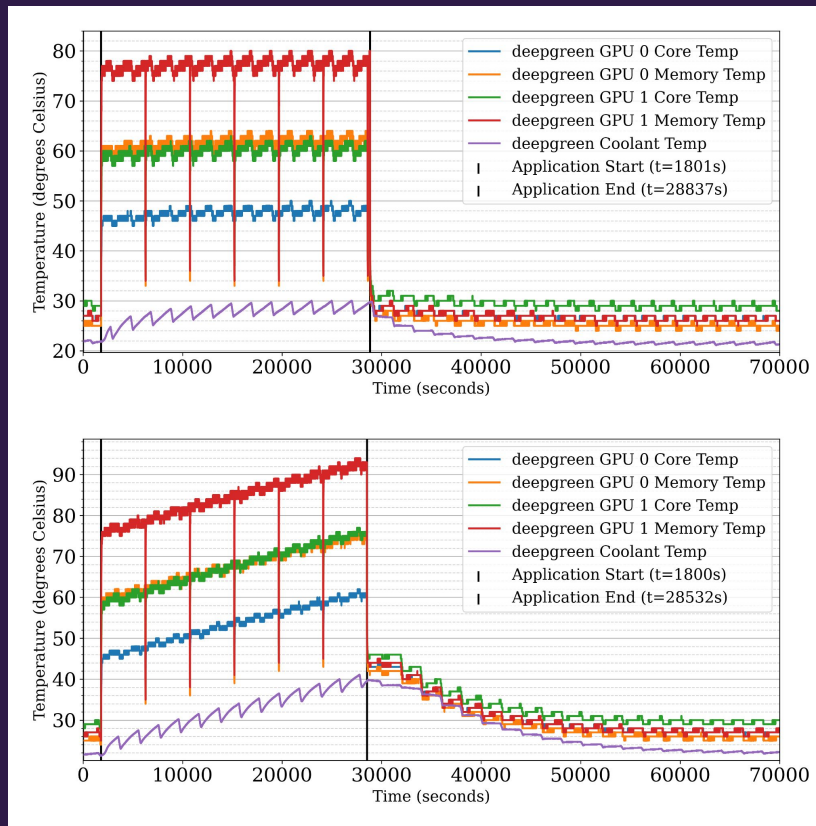


≡ Software Benchmarks & Metrics

- GPU-Accelerated Applications
 - Memory-Bottlenecks
 - EMOGI
 - Compute-Bottlenecks
 - CUDA DGEMM
 - MD5 Bruteforcer
- Multi-Node CPU Applications
 - Memory-Bottlenecks
 - NPB DT (Class C)
 - NPB IS (Class D)
 - Compute-Bottlenecks
 - NPB EP (Class E)
 - HPCC HPL
- CPU metrics
 - Core temperature and frequencies
- GPU metrics
 - Core and memory temperature and utilization
 - Power usage and performance state
- NVMe metrics
 - Temperature
- PDU metrics
 - Total system power draw
- SPLIC Tank metrics
 - Coolant and water temperature
 - Power consumption and pump RPM
 - Coolant and water flow rates
 - Effective dissipation and mPUE



Rising Tide Lifts All Boats Together



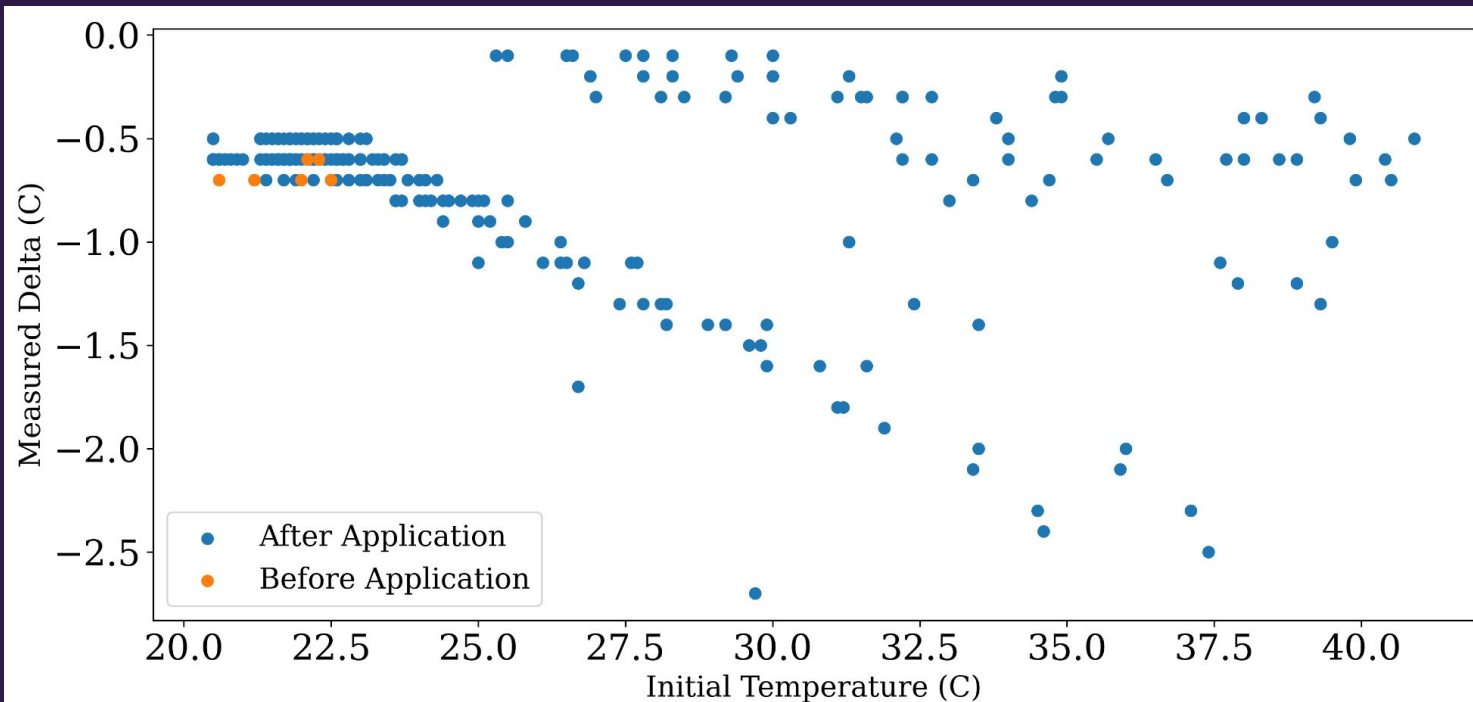
- Top: Normal SPLIC operation
- Bottom: No chilled water until application period ends
- Coolant temperature affects ALL components equally
 - Physical proximity to hot GPUs does not increase average temperature or rate of increase
 - Even within the same server node!

Hardware Component	Idle Temp. Min/Mean /Max (°C)	Interval (s) to Increase Temp by 1°C	Op. Temp. Observed Max (°C)
SPLIC Coolant	19.80 / 21.41 / 22.50	970.99	47.10
deepgreen CPU	14.75 / 17.51 / 19.66	702.20	55.62
deepgreen GPU	24.25 / 26.54 / 28.00	266.83	96.00
deepgreen NVMe	14.00 / 17.91 / 20.00	1068.48	44.00
n01 CPU	17.44 / 21.30 / 24.04	607.33	66.00
n02 CPU	17.67 / 20.81 / 24.26	560.35	69.00



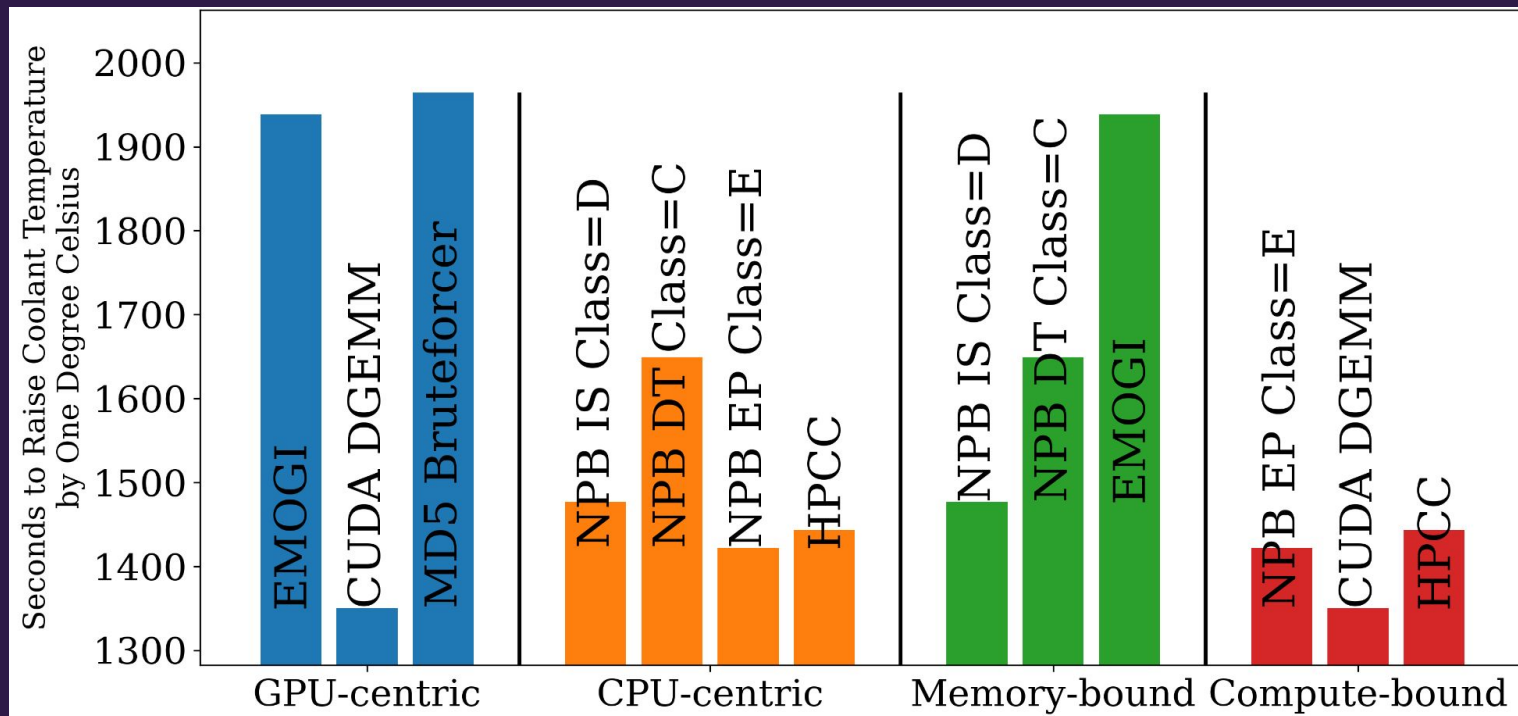
≡ Inconsistent But Improving Efficiency

- Pumps generally manage to lower coolant temperature better with greater differential
- However, many pumping periods end up being *ineffective*, even compared to initial conditions

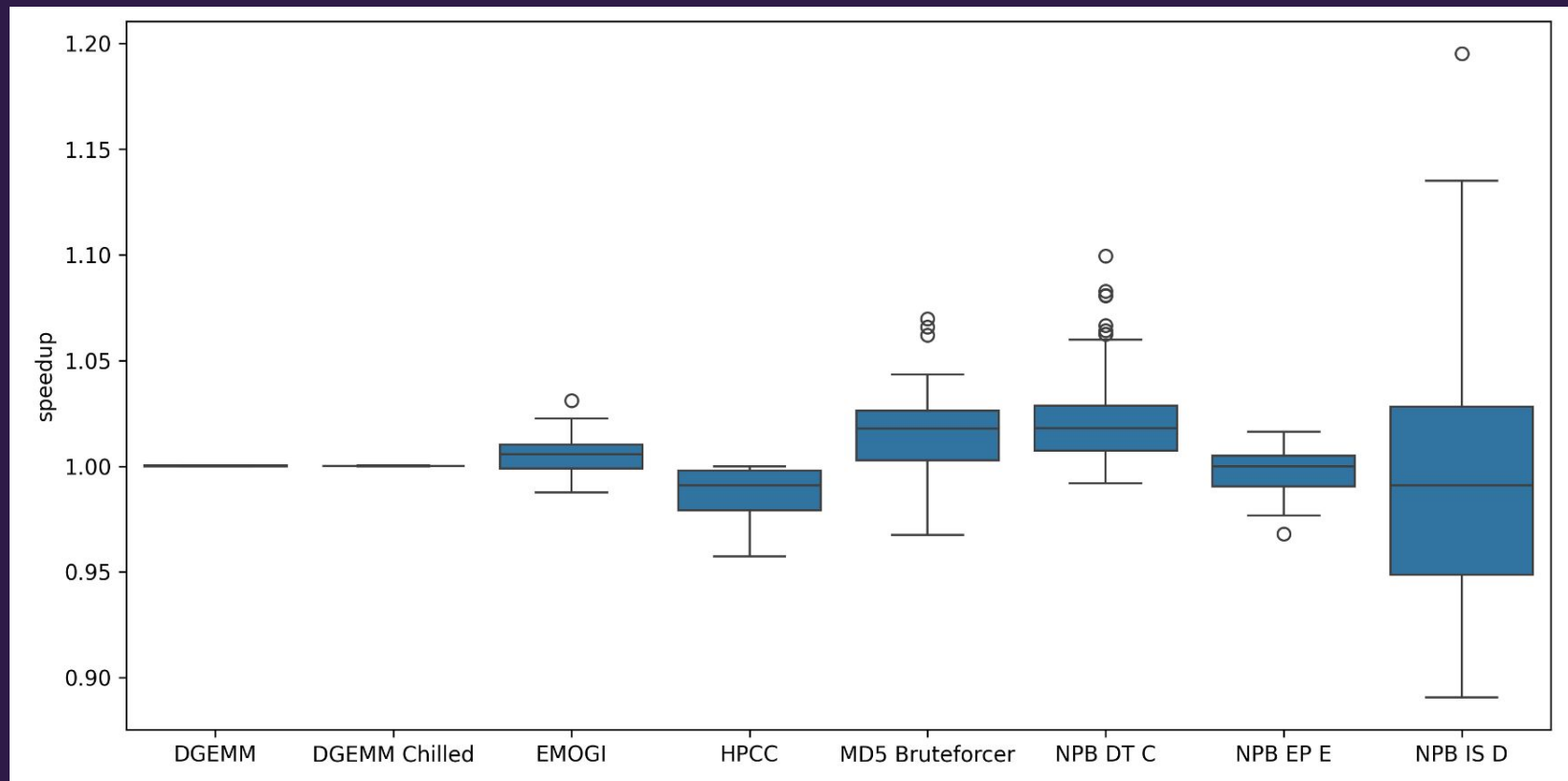


Compute-Utilization More Important than Device

- Lower values = faster heat accumulation



≡ No Significant Performance Deviations



Conclusions

- Coolant normalizes per-component heat increases
 - Circulation semi-sufficient due to excess coolant volume
- Delayed responses require thermal benchmarking
 - Can permit components to throttle at low coolant temperatures
- Compute is primary driver of heat accumulation
- External observations and considerations
 - Preparation and maintenance
 - Irreversible commodity hardware modifications
 - Additional care during maintenance
 - Reliability
 - No degradations in computing performance
 - Plastic components may require periodic replacement
 - Power outage lead to pump failure – requires specialized nontrivial maintenance



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Experiment code and data are available online

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